Hyperseed Formalizations: Orientation Note 0001

OmegaClaw / ZeroBot working notes for Ben Goertzel

2026-06-27

Abstract

This note seeds a public repository for incremental Hyperseed formalizations arising from OmegaClaw, ZeroBot, and Ben Goertzel's research interactions. It is not a claim of completed theory; it is a provenance-bearing scaffold for definitions, conjectures, proof sketches, and experiment formalizations.

1 Source orientation

The initial orientation sources are Ben Goertzel's public Hyperseed v2 article and the four linked PDF documents preserved in the OpenClaw research library on 2026-06-27:

- *Hyperseed ontology: formal presentation and theoretical development.*
- *Hyperseed: A Scrutability-Style Core Ontology and a Practical Inference-Control Scaffold for Probabilistic Logic Networks.*
- *Representing Scientific Experiments in MeTTa: SUMO/EXPO Structure with Hyperseed Epistemic Annotations.*
- *Metta-stasizing SUMO: A Typed MeTTa Encoding for Type Checking, PLN Reasoning, and Hyperseed Alignment.*

2 A minimal working frame

Definition 1 (Experienced research event)**.** *An* experienced research event *is a tuple*

$$E = \langle O, A, C, R, P \rangle$$

where O is an observing agent or agent-ensemble, A is an action or interaction, C is the operative context, R is the resulting observation or artifact, and P is provenance sufficient for later scrutiny.

Definition 2 (Hyperseed formalization act)**.** *A* Hyperseed formalization act *maps an experienced research event E into a structured object*

$$F(E) = \langle D, X, K, T, Q \rangle$$

where D is a set of candidate definitions, X examples and counterexamples, K conjectures, T theorem/proof-sketch material, and Q open questions or calibration requests.

Conjecture 1 (Experiment-to-ontology feedback)**.** *For a sustained research process, repeatedly applying F to concrete experiments and agent interactions can improve both experiment design and ontology design, provided the formalization records uncertainty, provenance, and failure modes rather than only polished successes.*

Example 1 (petta-chem as initial target)**.** *A deterministic candidate-selection run in `petta-chem` may be treated as an experienced research event whose context includes a PeTTa/MeTTa runtime, symbolic chamber state, candidate applicability rules, and observed post-tick abundance invariants. A first Hyperseed formalization should relate this to process, pattern, causality, attention or resource allocation, and experiment-evidence annotations.*

3 Near-term agenda

1. Formalize `petta-chem` experiment records using the scientific-experiment paper's SUMO/EXPO plus Hyperseed epistemic annotation style.
2. Formalize OmegaClaw's Telegram smoke test and idle-loop issue as agent-experience events involving communication, attention, resource use, and action gating.
3. Extract a small reusable notation for "interaction afterthoughts": immediate conversational response first, later Hyperseed reflection when useful.

4 Open questions for Ben

1. Should the repository prefer mathematically polished theorem/proof style, or a lab-notebook style that tolerates more speculative formal sketches?
2. Which Hyperseed primitives should be treated as the default starting vocabulary for formalizing software experiments?
3. How much MeTTa syntax should be paired with each LaTeX formalization?